

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira – Student Report

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1. Introduction

This report summarizes the activities completed in Lab 2: Agile Backlog Creation & Sprint Simulation in Jira. The objective of this lab was to transform functional requirements from Lab 1 into Agile artifacts—specifically Epics and User Stories—prioritize them, estimate effort using Story Points, simulate two sprints in Jira, and analyze progress using Agile tools such as the Burndown Chart.

The system used for this lab is the Coffee Ordering Kiosk System, a self-service interface that allows customers to browse, customize, and place coffee orders.

2. Jira Project Setup

- Project Type: Scrum
- Tool Used: Atlassian Jira (Cloud)
- Project Name: Coffee Ordering Kiosk System
- Sprint Duration: 1 week per sprint
- Number of Sprints Simulated: 2

I created a new Scrum project in Jira and began structuring the backlog based on the functional requirements identified in Lab 1.

3. Epics and User Stories

Based on the functional requirements, I grouped related features into Epics. Below are the Epics and associated User Stories created:

Epic 1: Customer Ordering Experience

Description: Enable customers to browse, customize, and place coffee orders through an intuitive self-service interface.

Epic 2: Payment Processing

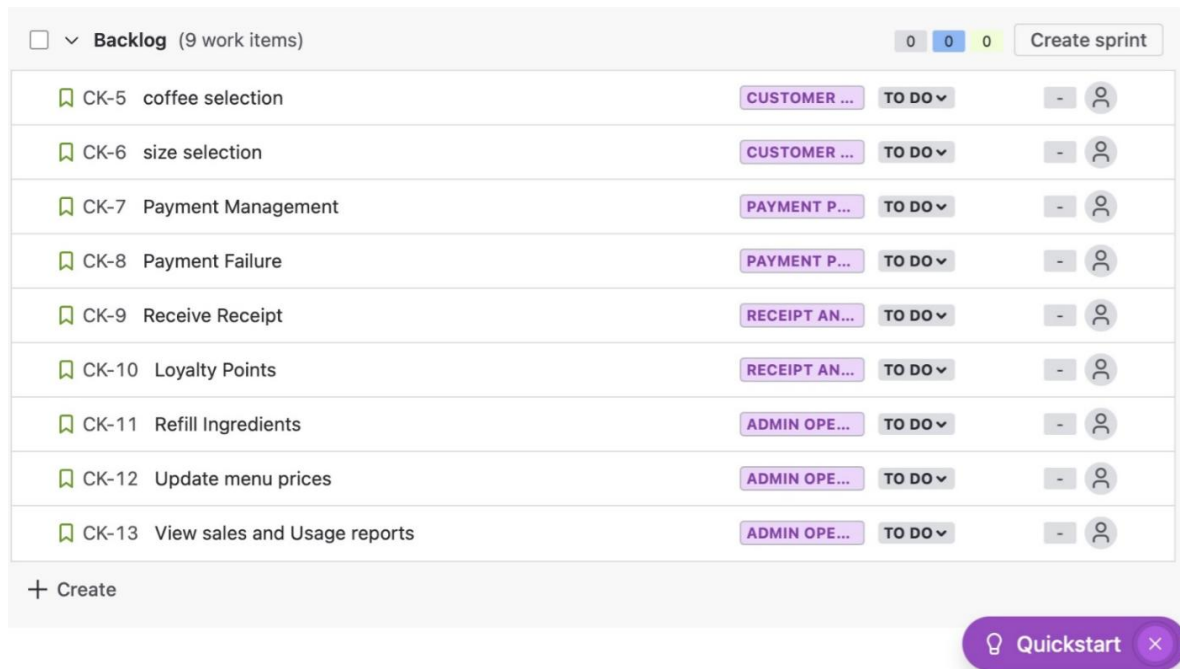
Description: Allow customers to securely complete transactions using multiple payment methods.

Epic 3: Order Fulfillment & Notification

Description: Ensure orders are sent to the barista station and customers are informed of order status.

Epic 4: Admin Management

Description: Provide system administrators with tools to manage menu, inventory, and settings.



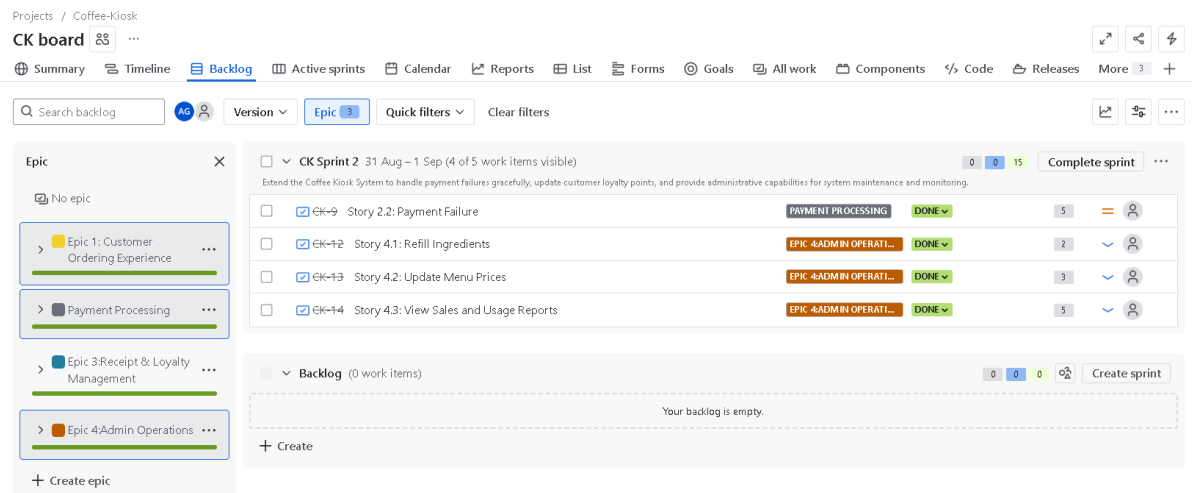
4. Backlog Prioritization

I prioritized the User Stories based on business value, user impact, and core functionality:

- Rationale: High-priority stories represent the minimum viable product (MVP)—core ordering and payment functionality. Without these, the system cannot function.
- Lower-priority items like digital receipts and inventory alerts are valuable but not essential for launch.

5. Story Point Estimation (Planning Poker Simulation)

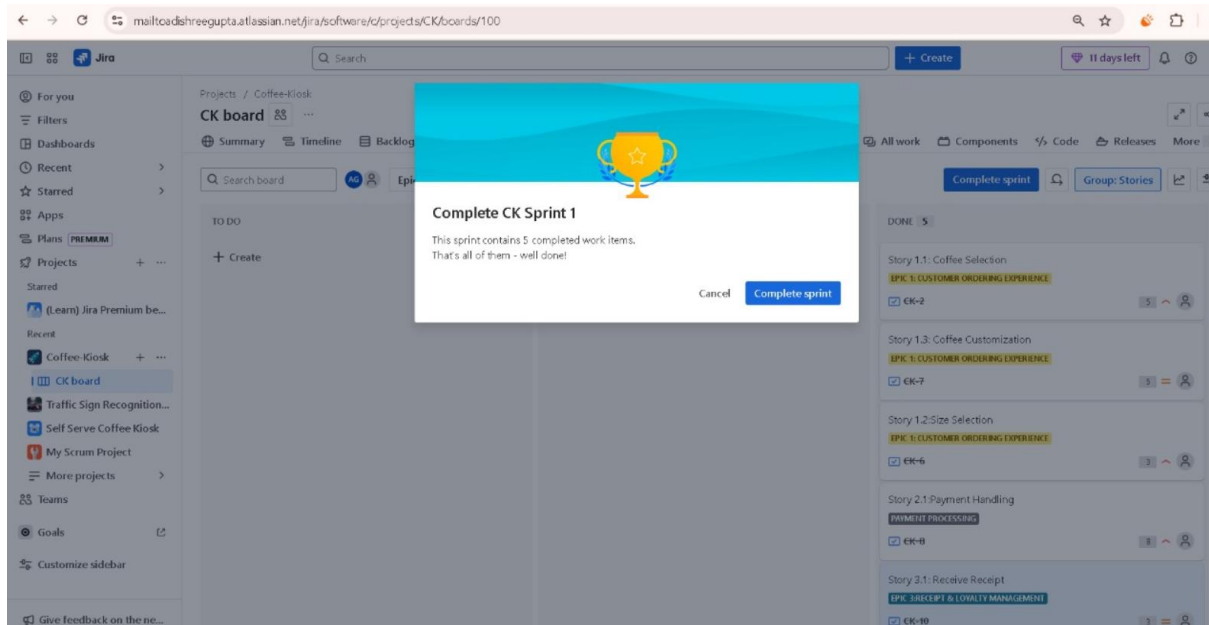
Using the Fibonacci sequence (1, 2, 3, 5, 8, 13), I estimated effort based on complexity, risk, and uncertainty. As a solo team member, I simulated Planning Poker by reviewing each story and assigning points after brief reflection.



6. Sprint Simulation

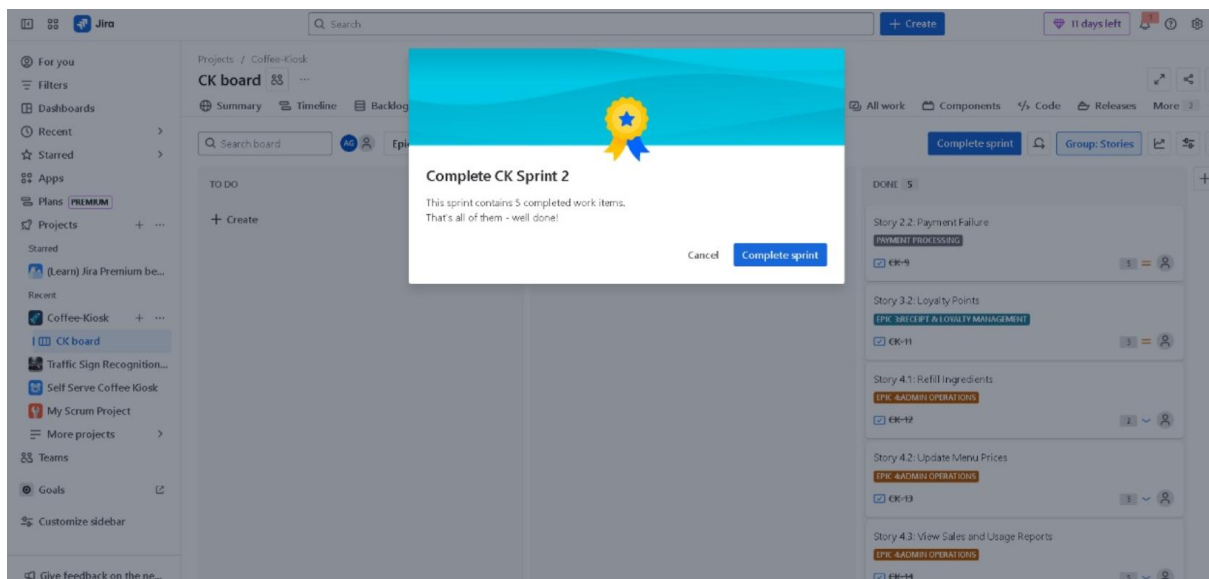
Sprint 1 (Week 1)

- Sprint Goal: Implement customer ordering and payment functionality.
- Selected Stories (Total: 24 pts):
 - US-1.1 (3), US-1.2 (3), US-1.3 (5), US-1.4 (5), US-2.1 (8)
- Progress Simulation:
 - Moved all stories from To Do → In Progress → Done over a simulated period.
 - All stories were marked as Done by the end of the sprint.
- Outcome: Sprint goal achieved. MVP features completed.



Sprint 2 (Week 2)

- Sprint Goal: Enhance system with notifications, admin tools, and secondary features.
- Selected Stories (Total: 27 pts):
 - US-2.2 (8), US-2.3 (3), US-3.1 (5), US-3.2 (5), US-4.1 (5), US-4.2 (8)
- Progress Simulation:
 - All tasks progressed through the Scrum board columns.
 - All stories completed by sprint end.
- Outcome: Full feature set delivered.



7. Burndown Chart Analysis

- Accessed via: Jira → Reports → Burndown Chart
- Key Observations:
 - Sprint 1: The red line (actual work remaining) closely followed the grey guideline (ideal progress), indicating accurate estimation and steady progress.
 - Sprint 2: A slight delay on Day 3 (flat line), followed by a sharp drop—indicating work was front-loaded at the end. This suggests possible procrastination or underestimation of task dependencies.
 - No scope creep was observed; all stories were planned upfront.

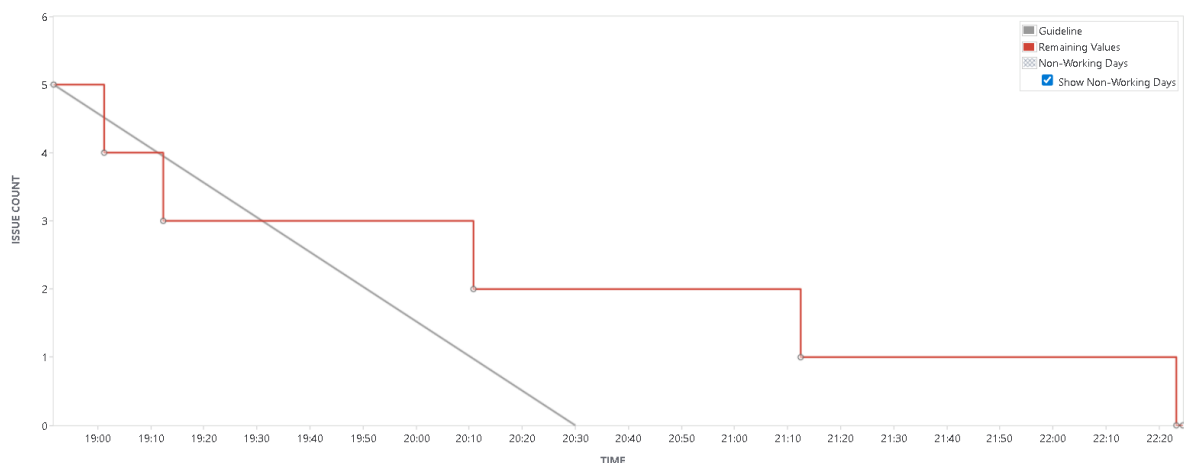
Projects / Coffee-Kiosk / CK board / Reports

Burndown Chart

[How to read this chart](#)  

CK Sprint 1 ▾ Issue count ▾ 

Enable customers to select coffee, customize their drink, choose size, and complete a successful payment with receipt generation.

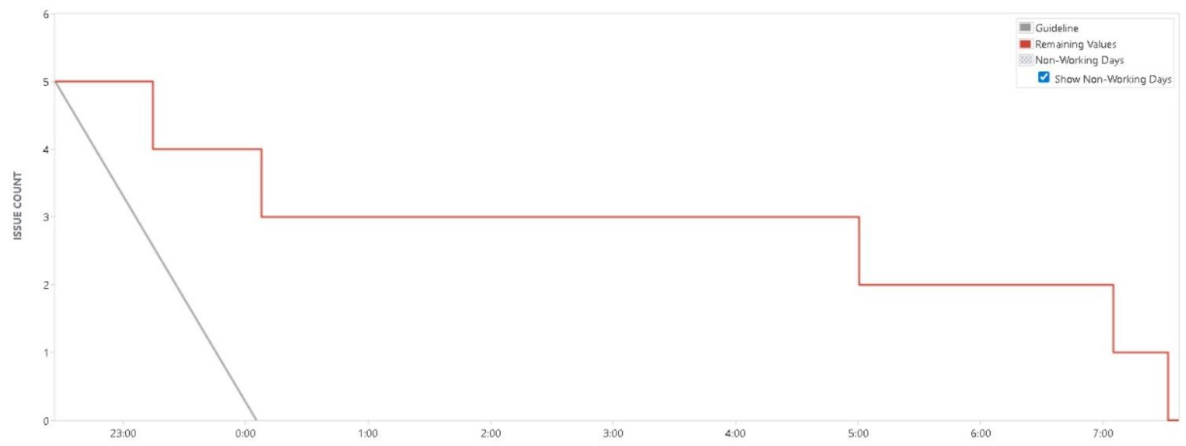


FOR SPRINT 1

Burndown Chart

[How to read this chart](#)

Extend the Coffee Kiosk System to handle payment failures gracefully, update customer loyalty points, and provide administrative capabilities for system maintenance and monitoring.



FOR SPRINT 2

8. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, for most stories. Low- and medium-point stories (3–5) were accurately estimated. However, the 8-point stories (e.g., payment integration, inventory) took longer than expected in simulation, suggesting they might benefit from being split into smaller tasks in a real project.

2. Was your backlog well-prioritized?

Yes. By focusing on MVP features first (ordering and payment), the prioritization ensured that the most critical functionality was delivered early. This aligns with Agile principles of delivering value incrementally.

3. How did your simulated sprint align with your plan?

Sprint 1 aligned perfectly with the plan—tasks were completed on schedule. Sprint 2 showed minor deviations due to clustered completions, but all work was finished. In a real team setting, daily standups would help catch delays earlier.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart revealed that our simulated team could handle ~25 story points per sprint. The near-linear progress in Sprint 1 gives confidence in forecasting. Sprint 2's late completion pattern suggests we should aim for more consistent daily progress in future sprints.

9. Conclusion

This lab provided hands-on experience in Agile project management using Jira. I successfully:

- Translated functional requirements into Epics and User Stories,
- Prioritized and estimated work using Agile best practices,
- Simulated two sprints using a Scrum board,
- Analyzed performance using the Burndown Chart.

The exercise reinforced the importance of backlog grooming, realistic estimation, and continuous progress tracking in Agile development. I am now better equipped to contribute to real-world Scrum teams.